

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 3

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1 Introduction

Following the theoretical study, dataset exploration, and project planning in the first two weeks, Week 3 marks the transition from research to active code implementation. The focus this week was on establishing the software foundation for the project, primarily centered around dataset loading, image preprocessing, and the initial setup for feature extraction using classical Digital Image Processing (DIP) libraries in Python (e.g., OpenCV, scikit-image).

2 Dataset Setup and Organization

To efficiently handle the AITEX dataset, a custom data loader module was implemented. This module automatically traverses the directory structure to pair the high-resolution fabric images with their corresponding ground-truth defect masks.

An initial dataset split was established to support both parameter tuning and final evaluation, using an approximate ratio of **60:20:20**:

- **Training Set (60%)**: Defect-free images used to characterize the baseline normal texture profile for each fabric type.
- **Validation Set (20%)**: A held-out set of mixed images used to tune and compare the fixed vs. adaptive parameter strategies without touching the test data.
- **Test Set (20%)**: Defective images paired with pixel-level ground-truth masks, reserved exclusively for final quantitative evaluation (e.g., IoU, pixel accuracy).

3 Image Preprocessing Pipeline

Based on our conceptualization, a preliminary data flow pipeline was devised for the detection system:

The raw images in the AITEX dataset are exceptionally large (4096×256 pixels). Processing these images directly using computationally intensive texture descriptors (like GLCM and Gabor filters) across multiple scales is inefficient.

To address this, an initial preprocessing pipeline was developed:

- **Grayscale Conversion**: Ensuring all images are processed as single-channel grayscale, as texture is primarily a structural property rather than a color property.
- **Patch Extraction**: Given the extreme 4096×256 aspect ratio, images are divided into 16 non-overlapping 256×256 square patches. This maintains the structural integrity of the woven threads while providing manageable, uniform inputs for the texture descriptors without overwhelming system memory.
- **Normalization**: Histogram equalization techniques are being explored to normalize contrast variations caused by uneven lighting during the fabric capture process.

4 Implementation of Texture Descriptors

With the preprocessing pipeline taking shape, initial code structures for extracting texture descriptors were established.

4.1 Local Binary Patterns (LBP)

The implementation of the LBP descriptor was initiated using the `scikit-image` library. Functions were written to compute the LBP representation of the normal fabric patches. The resulting histograms capture the local spatial patterns, which will serve as the reference feature vector for identifying anomalies.

4.2 GLCM Setup

Preliminary setup for the Gray-Level Co-occurrence Matrix (GLCM) was also started. We defined the initial parameters (distances and angles) to capture the directional nature of the woven threads, which will be used to extract statistical features like contrast, energy, and homogeneity in the upcoming weeks.

5 Work Completed This Week

- Set up the Python development environment and imported necessary classical DIP libraries (OpenCV, scikit-image).
- Implemented the dataset loading module to pair images with ground-truth masks.
- Designed and began implementing the image preprocessing pipeline (grayscale conversion, normalization, and patch extraction).
- Established the initial training/testing data split.
- Wrote the initial code to extract Local Binary Patterns (LBP) histograms from normal fabric textures.

6 Plan for Next Week

- Finalize the image preprocessing pipeline and determine optimal patch sizes for texture analysis.
- Complete the implementation of GLCM feature extraction.
- Begin implementing Gabor filter banks for frequency-domain texture analysis.
- Develop the comparison logic to measure the distance/difference between test image features and normal training features to generate preliminary defect probability maps.

End of Week 3 Report

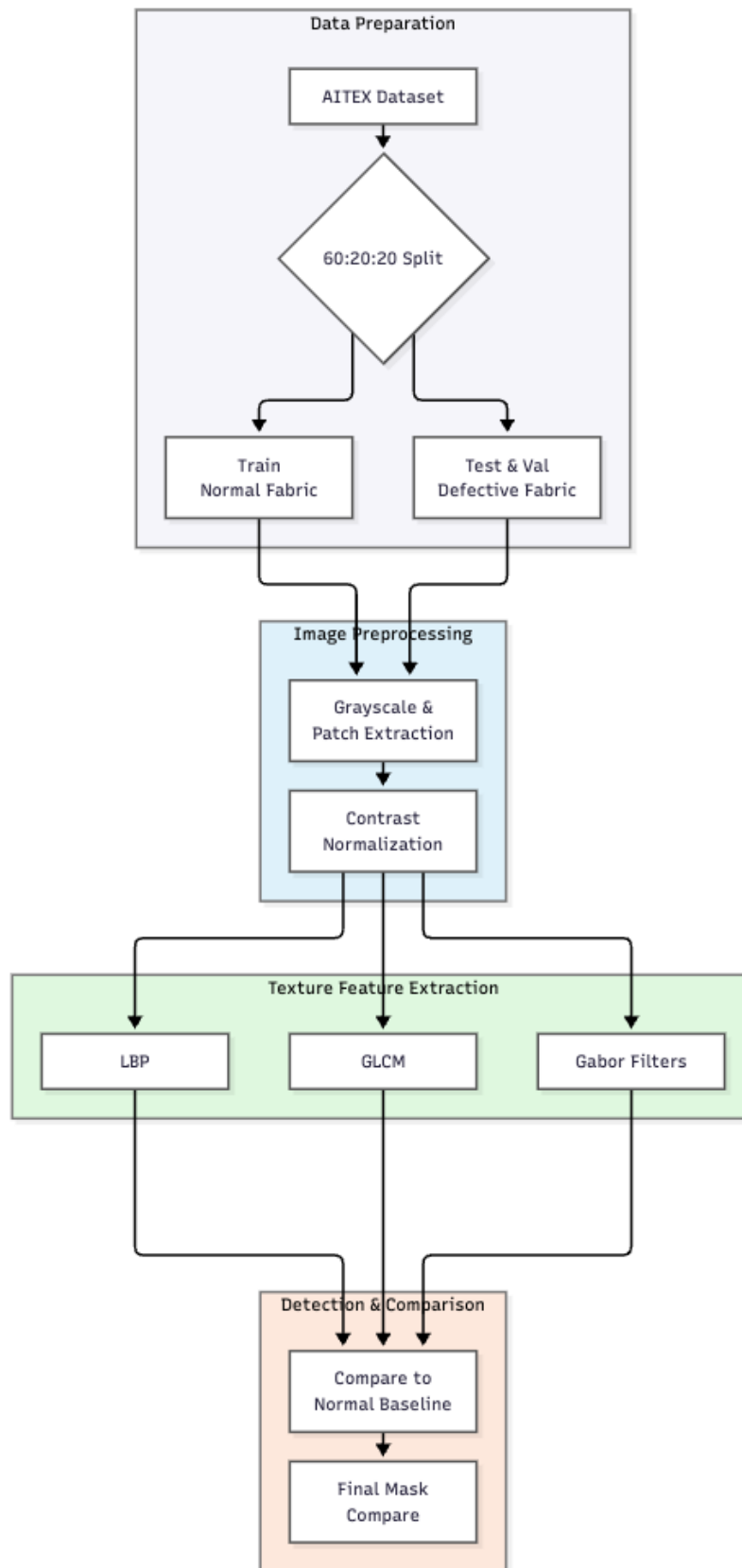


Figure 1: Proposed Initial Pipeline for Defect Detection